

1. Hello.c

```
#include<stdio.h>
```

```
int main()
{
    printf("Hello Linux\n");
    return 0;
}
```

Compile: gcc hello.c -o hello

./hello

```
vivek@DESKTOP-4VP07FT:~$ gcc hello.c -o hello
vivek@DESKTOP-4VP07FT:~$ ./hello
Hello Linux
vivek@DESKTOP-4VP07FT:~$ |
```

2. Read User Name

```
#include<stdio.h>
```

```
int main()
{
    char name[30];

    printf("Enter your name: ");
    scanf("%s",name);

    printf("Welcome %s\n",name);

    return 0;
}
```

```
vivek@DESKTOP-4VP07FT:~$ nano username.c
vivek@DESKTOP-4VP07FT:~$ gcc username.c -o username
vivek@DESKTOP-4VP07FT:~$ ./username
Enter your name: vivek
Welcome vivek
vivek@DESKTOP-4VP07FT:~$ |
```

3. Display Process ID

```
#include<stdio.h>
#include<unistd.h>

int main()
{
    printf("Process ID = %d\n",getpid());

    return 0;
}
```

```
vivek@DESKTOP-4VP07FT:~$ nano pid.c
vivek@DESKTOP-4VP07FT:~$ gcc pid.c -o pid
vivek@DESKTOP-4VP07FT:~$ ./pid
Process ID = 2747
vivek@DESKTOP-4VP07FT:~$ |
```

4. Fork()

```
#include<stdio.h>
#include<unistd.h>

int main()
{
    fork();

    printf("Hello\n");

    return 0;
}
```

```
vivek@DESKTOP-4VP07FT:~$ nano fork.c
vivek@DESKTOP-4VP07FT:~$ gcc fork.c -o fork
vivek@DESKTOP-4VP07FT:~$ ./fork
Hello
Hello
vivek@DESKTOP-4VP07FT:~$ |
```

5. Parent and Child

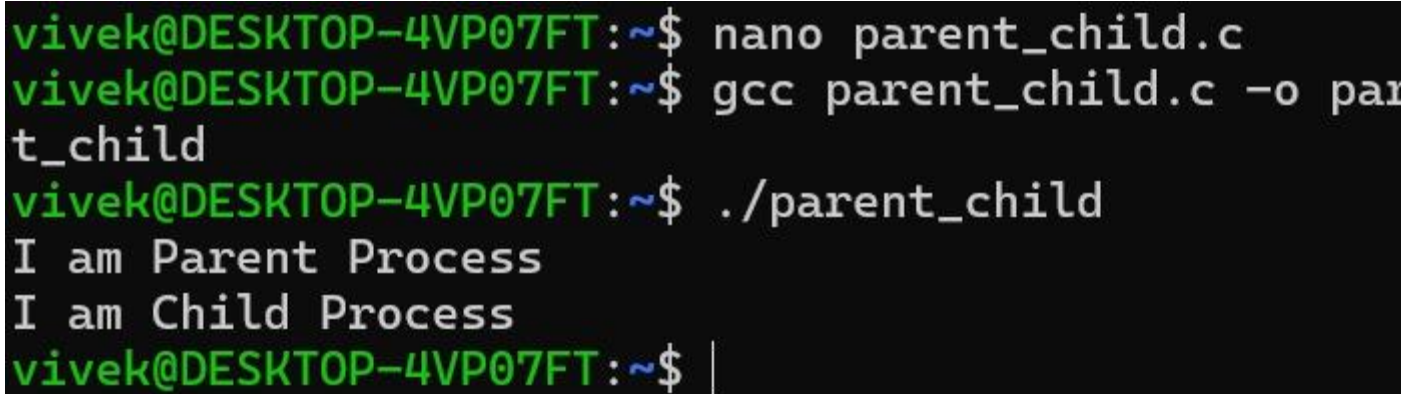
```
#include<stdio.h>
#include<unistd.h>

int main()
{
    pid_t pid;

    pid=fork();

    if(pid==0)
    {
        printf("I am Child Process\n");
    }
    else
    {
        printf("I am Parent Process\n");
    }

    return 0;
}
```

A terminal window with a black background and green text. The prompt is 'vivek@DESKTOP-4VP07FT:~\$'. The user enters 'nano parent_child.c', then 'gcc parent_child.c -o parent_child', and finally './parent_child'. The output shows 'I am Parent Process' followed by 'I am Child Process' on the next line. The prompt returns.

```
vivek@DESKTOP-4VP07FT:~$ nano parent_child.c
vivek@DESKTOP-4VP07FT:~$ gcc parent_child.c -o parent_child
vivek@DESKTOP-4VP07FT:~$ ./parent_child
I am Parent Process
I am Child Process
vivek@DESKTOP-4VP07FT:~$ |
```

6. Display parent and Child PID

```
#include<stdio.h>
#include<unistd.h>

int main()
{
    pid_t pid;

    pid=fork();

    if(pid==0)
    {
        printf("Child PID = %d\n",getpid());
    }
}
```

```

        printf("Parent PID = %d\n",getppid());
    }
    else
    {
        printf("Parent PID = %d\n",getpid());
    }

    return 0;
}

```

```

vivek@DESKTOP-4VP07FT:~$ nano pid_display.c
vivek@DESKTOP-4VP07FT:~$ gcc pid_display.c -o pid_
splay
vivek@DESKTOP-4VP07FT:~$ ./pid_display
Parent PID = 2892
Child PID = 2893
Parent PID = 532
vivek@DESKTOP-4VP07FT:~$ |

```

7. Wait()

```

#include<stdio.h>
#include<unistd.h>
#include<sys/wait.h>

int main()
{
    pid_t pid;

    pid=fork();

    if(pid==0)
    {
        printf("Child Running\n");
    }
    else
    {
        wait(NULL);

        printf("Parent Running After Child\n");
    }

    return 0;
}

```

```
vivek@DESKTOP-4VP07FT:~$ nano wait.c
vivek@DESKTOP-4VP07FT:~$ gcc wait.c -o wait
vivek@DESKTOP-4VP07FT:~$ ./wait
Child Running
Parent Running After Child
vivek@DESKTOP-4VP07FT:~$ |
```

8. Wait()

```
#include<stdio.h>
#include<unistd.h>

int main()
{
    printf("Before exec()\n");

    execlp("ls","ls",NULL);

    printf("After exec()\n");

    return 0;
}
```

```
vivek@DESKTOP-4VP07FT:~$ nano exec.c
vivek@DESKTOP-4VP07FT:~$ gcc exec.c -o exec
vivek@DESKTOP-4VP07FT:~$ ./exec
Before exec()
S2Commands    fork        hello.c.save  pid.c
ex.c.save     fork.c      parent_child  pid_display
exec          hello       parent_child.c pid_display.c
exec.c        hello.c     pid           username
vivek@DESKTOP-4VP07FT:~$ |
```

9. fork() + exec()

```
#include<stdio.h>
```

```
#include<unistd.h>
#include<sys/wait.h>

int main()
{
    pid_t pid;

    pid=fork();

    if(pid==0)
    {
        execlp("date","date",NULL);
    }
    else
    {
        wait(NULL);

        printf("Child Completed\n");
    }

    return 0;
}
```

```
vivek@DESKTOP-4VP07FT:~$ nano fork_exec.c -o fork
Invalid operating directory: fork_exec
vivek@DESKTOP-4VP07FT:~$ |
```